

[< Corn](#)**AGRICULTURE**

Armyworms

Armyworm: *Mythimna* (= *Pseudaletia*) *unipuncta*

Beet armyworm: *Spodoptera exigua*

Western yellowstriped armyworm: *Spodoptera praefica*

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Description of the Pest

Eggs of the armyworm and beet armyworm are pale green to pink, striated, and deposited in a mass that is covered with a white cottony material. At first, the tiny, newly hatched larvae feed in a colony, skeletonizing the first few leaves on which they feed. The larvae are about 1.25 inches long when full grown. Armyworms are greenish

brown in color with several longitudinal stripes, while [beet armyworm](#) are mottled olive green to almost black.

[Eggs](#) of the western yellowstriped armyworm are similar to those of the beet armyworm. The egg masses are larger and covered by a gray cottony material. Newly hatched larvae also feed in colonies; [mature larvae](#) attain a length of 1.5 to 2.0 inches and are black with a prominent yellowish stripe and several narrow bright ones on each side of the body. An intense black spot is usually visible on each side of the first legless segment behind the head.

Damage

Newly hatched armyworms feed in colonies and skeletonize leaves. Larvae that are half grown or more will feed singly on leaves, in the ear, or on the tassels. Damage by armyworm to the ears resembles that caused by corn earworm. Because of the damage to tassels and ears, tolerance for armyworm damage in sweet corn is very low.

Management

Armyworms are often attracted to fields with [barnyardgrass](#). When they are done feeding on barnyardgrass, larvae move to corn so it is important to control this weed. Specific treatment thresholds have not been established for armyworms on field corn, but treatment is seldom necessary. Sweet corn, however, has a greater potential for damage and may require treatment. While these pests may be present any time from June through September, populations are usually most damaging in late summer. For sweet corn, pheromone traps may be useful to determine the timing of moth flights and subsequent larval infestations. In those rare instances where control measures are required, the beet armyworm is more difficult to control than the western yellowstriped armyworm and may require the higher treatment rate of the materials recommended below.

Insecticide applications will be most effective if applied against small larvae. Frequently, spot treatments are sufficient.

Organically Acceptable Methods

Controlling weeds, especially barnyardgrass, and applying sprays of *Bacillus thuringiensis* are organically acceptable management tools.

Pesticides and Natural Enemy Releases

Not all registered pesticides are listed. **The following are ranked by their IPM value, with the most effective and least harmful to natural enemies, honey bees, and the [environment](#) listed at the top of the table.** When choosing a pesticide, consider information related to water and [air quality](#), resistance management, and the pesticide's properties and application timing. Use [PestManage](#) to compare summarized management options for different pests in the same crop. Always carefully read the label of the product being used and take all necessary [precautions when handling pesticides](#).



[Expand table](#)

Rank	Active ingredient	Example t
Application Timing Varies (See <i>UC IPM Pest Management</i>		
A	indoxacarb	Avaunt
B	methoxyfenozide	Intrepid 2F

Rank ▲	Active ingredient ▲	Example t
C	methomyl [*] —	Lannate SF
D	Bacillus thuringiensis ssp. aizawai	various pro
E	spinetoram	Radiant SC
F	esfenvalerate	Asana XL
G	permethrin	Pounce 3.2

— No information.

I

Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.

II

Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to contaminate water accessible to bees including puddles.

III

No bee precaution, except when required by the pesticide label or regulations.

- ‡ [a](#) [b](#) Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest

interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.

- * [a](#) Permit required from county agricultural commissioner for purchase or use.
- # [↩](#) Acceptable for use on certified organic crops. Check with your certifier to confirm before application.
- ¹ [↩](#) Group numbers for insecticides and miticides are assigned by the [Insecticide Resistance Action Committee](#) (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B; insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.
- ² [↩](#) Range of insect and mite groups affected by a pesticide. **Broad** means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
- ³ [↩](#) Risk of harm to honey bees. For more information, see [Bee Precaution Pesticide Ratings](#).
- ⁴ [↩](#) Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.
- ⁵ [a b](#) Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on

factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.

- ⁶ [↩](#) Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- ⁷ [↩](#) Risk of harm to fish from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ⁸ [↩](#) Risk of harm to fish from adsorbed runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ⁹ [↩](#) Risk of harm to fish from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹⁰ [↩](#) Risk of harm to humans from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹¹ [↩](#) Risk of harm to humans from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹² [↩](#) Date information was last updated in the *UC IPM Pest Management Guidelines*.
- ¹³ [↩](#) Toxic to some natural enemies (lacewing and syrphid fly larvae, predatory beetles, and thrips) when sprayed and up to 5 to 7 days after.
- ¹⁴ [↩](#) Residual is moderate if solution is between a pH of 7 and 8.



UC IPM Pest Management Guidelines: Corn

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